

Parth Agrawal

+91-8269432208

parthagrawal4675@gmail.com

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Research Profile

- B.Tech. student in Information Technology at IIIT Allahabad with a minor in Entrepreneurship and a CGPA of **9.24/10**, with research interests spanning Machine Learning, Artificial Intelligence, Cybersecurity, and Privacy.
- Interested in problems where information is deliberately hidden or difficult to observe, particularly how machine learning can recover meaningful and generalizable structure from observable signals without requiring direct access to sensitive content.
- Current research centers on encrypted DNS traffic intelligence, with emphasis on feature representation, experimental design, robustness, generalization, interpretability, reproducibility, and the boundary between encryption-based privacy and residual traffic observability.

Education

Indian Institute of Information Technology, Allahabad

Aug 2023 – Present

- BTech in Information Technology (Major) and Entrepreneurship (Minor) | CGPA: **9.24/10**

Christ Church Boys' Senior Secondary School

2019 – 2022

- CBSE: **95.3%** | CBSE (PCM): **92.8%**

Research Experience

Encrypted DNS Traffic Intelligence | *Machine Learning, Network Security, Encrypted Traffic Analysis* | 2026 – Present

- Developing an ML-based experimental framework to investigate whether **encrypted DNS traffic remains identifiable or classifiable** without accessing plaintext DNS payloads, treating the problem as a study of residual information exposed by encrypted communication.
- Investigating **DNS-over-HTTPS (DoH)**, **DNS-over-TLS (DoT)**, and **DNS-over-QUIC (DoQ)** and characterizing observable traffic behavior through flow statistics, packet metadata, timing relationships, burst structure, connection behavior, and protocol-level features.
- Constructing structured datasets for controlled training and testing, defining traffic-level feature representations, and developing feature-engineering pipelines that transform packet/flow observations into model-ready representations while preserving the encrypted-payload constraint.
- Building supervised learning, evaluation, and visualization workflows to compare classification behavior across experimental settings and to determine whether apparently informative traffic patterns persist beyond a single capture environment or dataset.
- Studying **feature importance, robustness, generalization, and dataset/environment-specific artifacts**, with the research goal of understanding what models actually learn rather than treating predictive accuracy alone as evidence of a meaningful finding.
- Developing an interactive research platform for dataset management, experiment configuration, model evaluation, and result visualization to support repeatable experimentation and systematic comparison of traffic characteristics.
- Research direction: **privacy-preserving traffic analysis, encrypted protocol fingerprinting, network ML, feature representation, reproducibility, and the privacy-observability boundary.**

ML-Based Tor Traffic Analysis | *Machine Learning, Network Security, Traffic Classification*

- Explored ML-based classification and analysis of **Tor network traffic** using network-level flow and statistical characteristics rather than relying on application payloads, providing an early exposure to learning under encryption and anonymity constraints.
- Examined traffic-flow representations and statistical characteristics as potential signals for classification, while considering the effects of variable traffic behavior, limited observability, noise, and changing network conditions.
- Studied the difficulty of separating **generalizable traffic structure from incidental correlations**, which motivated a broader interest in robustness, experimental controls, and careful evaluation of ML models for privacy-preserving network analysis.
- This work forms part of the research trajectory leading to the current encrypted-DNS project: from studying ML on anonymized traffic toward investigating what encrypted protocols continue to reveal through observable communication patterns.

Research Methods & Interests

- **Machine Learning for Network Intelligence:** Supervised learning, feature engineering, traffic classification, model evaluation, feature importance, robustness, and generalization.
- **Encrypted & Privacy-Preserving Systems:** Encrypted DNS, DoH, DoT, DoQ, Tor traffic, encrypted traffic analysis, anonymity networks, and residual traffic observability.
- **Research Methodology:** Structured dataset construction, controlled experimentation, reproducible evaluation, visualization, interpretation of learned signals, and investigation of dataset/environment-specific artifacts.
- **AI Systems:** LLMs, grounded AI, multi-agent systems, intent routing, context management, behavioral analytics, and reliable integration of AI models into real-world systems.

Work Experience

Visa Inc. | *Software Engineering Intern*

May 2026 – Jul 2026

- Developed and published a production-grade React 19 + TypeScript component library (@vap/feediq-tool) adopted by Visa's Acceptance Platform, enabling reusable analytics dashboards and modular UI architecture.
- Engineered scalable merchant feedback analytics using **TanStack Query**, parallel batch loading, intelligent caching with Redis, and ClickHouse-backed APIs, reducing network overhead while improving dashboard responsiveness.
- Built interactive data visualizations including sentiment analytics, contribution heatmaps, funnel charts, bubble charts, and comparison dashboards using **Recharts**, while implementing comprehensive testing with **Vitest** and React Testing Library to ensure production reliability.

Selected AI / ML Projects

DailyForge – Grounded Multi-Agent AI Personal Operating System | [🔗](#)

2026

- Built a grounded multi-agent AI system combining **behavioral telemetry**, **goal planning**, **habit analytics**, **time-blocking**, and **AI-driven coaching**, with the objective of turning structured user data into context-aware actions rather than generic chatbot responses.
- Designed an **intent-routing architecture** that directs requests to specialized agents for habit coaching, schedule optimization, goal strategy, recovery analysis, and N-of-1 behavioral experiments.
- Grounded AI reasoning in structured historical telemetry including habits, execution snapshots, goals, energy patterns, and friction logs, enabling recommendations to be derived from observed behavior and longitudinal context.
- Developed analytical scoring and recommendation workflows, including a deterministic multi-factor behavioral score, and connected model outputs to actionable planning and scheduling operations.

Manobala – AI-Integrated Mental Health Support & Community Platform | [🔗](#) | [🔗](#)

- Developed an AI-enabled social support platform integrating a **Gemini-powered conversational assistant** with community support, expert consultation, contextual chat history, and role-based administrative workflows.
- Implemented conversational AI capabilities using the Gemini API while maintaining application context and integrating AI assistance with non-AI user workflows rather than treating the system as a standalone chatbot.
- Built the platform using **React**, **Node.js**, **Express**, **MongoDB**, and **Gemini API**, with role-aware workflows for users, experts, and administrators and supporting community interaction features.

QuickCart AI – AI-Powered Shopping Co-Pilot | [🔗](#)

- Built an AI-powered quick-commerce platform using **Next.js**, **TypeScript**, **AWS DynamoDB**, and **Ollama (Mistral 7B)** that converts natural-language shopping missions into structured and optimized product baskets.
- Designed an intent-understanding and basket-generation workflow connecting natural-language goals to context-aware product recommendations and personalized shopping actions.
- Implemented explainable recommendation outputs with confidence information and persistent cart state to connect model-driven decisions with an end-to-end user workflow.

YatriSewa – Platform for Maha Kumbh Mela 2025, Prayagraj | [🔗](#)

- Developed a full-stack platform for the Maha Kumbh Mela ecosystem integrating booking, payment, and emergency-assistance workflows with a focus on high-volume user interaction.
- Implemented an SOS-oriented emergency workflow and transactional booking functionality, applying full-stack engineering to a large-scale public-service use case.

Relevant Coursework

- **A+ Courses:** Introduction to Machine Learning (IT3002), Artificial Intelligence (IT3004), Cyber Security (IT3001), Image and Video Processing (IT3003), Data Visualization (IT5516).
- **Additional Foundations:** Data Structures and Algorithms, Design and Analysis of Algorithms, Computer Networks, Database Management Systems, Operating Systems, Software Engineering, Object-Oriented Methodologies.

Research Engagement

- **GDM APAC Research Symposium 2026:** Research-focused applicant interested in machine learning, AI, cybersecurity, privacy, and developing rigorous methodology for encrypted-DNS traffic research.

Academic and Extracurricular Achievements

- Recognized among the Top 1500 performers in **The Big Code 2026** from a highly competitive applicant pool.
- **LeetCode:** Achieved a maximum contest rating of **1913**, solved **783+** coding problems, and secured **Rank 268** (Top 4.18%) in **LeetCode Biweekly Contest 185**.
- **Participated in Google Solutions Challenge (GSC) 2024:** Engaged in innovative problem-solving and technology challenges.
- **Completed Workshops – FOSS Wing & Design Wing:** Gained hands-on experience with open-source technologies and foundational design principles.
- **Active Hackathon Contributor:** Engaged in **5+ national hackathons** including Amazon Hackon, Hackout, OOC, and Adobe.
- **Honoured with Gold Medal and Appreciation Certificate in IMO, IEO, and NSO** in 2022 and secured a Zonal Rank of 5 in IMO.

Coding Profiles

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| • Leetcode 
Max rating: 1913 | • Codechef 
Max rating: 1534 | • Codeforces 
Max rating: 1255 |
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Skills

Languages & Tools: C | C++ | Python | HTML | CSS | JavaScript | TypeScript | Git
AI / ML: Machine Learning | Deep Learning | LLMs | Multi-Agent Systems | Feature Engineering | Model Evaluation
Technologies & Frameworks: React.js | Next.js | Node.js | Express.js | MongoDB
Data & Infrastructure: SQL | Redis | ClickHouse | AWS DynamoDB | Docker | Kubernetes
Cybersecurity: Encrypted Traffic Analysis | Network Traffic Classification | Network Protocols | Cybersecurity

Leadership & Positions of Responsibility

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| Coordinator – Core Team, Aparoksha Technical Fest (IIIT Allahabad)
• Collaborated with multiple teams to ensure smooth management of technical events, hospitality, and participant engagement. | Mar 2026 – Present |
| Coordinator – Gravity, Blockchain Wing (IIIT-A, Prayagraj)
• Led coordination of workshops and speaker sessions focused on blockchain innovation and applications. | Sep 2025 – Present |
| Coordinator – Wellness Studio (IIIT-A, Prayagraj)
• Managed wellness initiatives and sessions aimed at enhancing student mental health and well-being on campus. | Aug 2025 – Apr 2026 |
| Member – Nirmiti, Fine Arts Society (IIIT-A, Prayagraj)
• Spearheaded art event design and organization for campus cultural festivals. | Sep 2023 – Sep 2024 |